

Ryan Law

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EDUCATION

University of California, Irvine <i>Incoming Transfer, B.S. in Mechanical Engineering</i>	Expected Fall 2026 Irvine, CA
Mt. San Antonio College <i>Mechanical Engineering Transfer Preparation</i>	2024 – 2026 Walnut, CA
• Relevant Coursework: Statics, Dynamics, Mechanics of Materials, MATLAB, Engineering Graphics	

EXPERIENCE

UCLA California NanoSystems Institute (CNSI) <i>Microfabrication and Nanopatterning Programs</i>	Oct. 2025 – Dec. 2025 Los Angeles, CA
- Fabricated wafer-based microstructures using photolithography, thin-film deposition, wet etching, and electron-beam lithography in cleanroom environments - Optimized fabrication workflow by evaluating resist thickness, exposure conditions, and process sequencing to improve pattern quality and repeatability - Characterized fabricated features under microscopy and used process observations to refine subsequent wafer iterations - Coordinated team task division during cleanroom processing to improve efficiency while maintaining contamination-control and safety standards	

Mt. San Antonio College VEX Robotics Team <i>Team Captain</i>	Dec. 2024 – Dec. 2025 Walnut, CA
- Led mechanical design of a collegiate VEX robot for a 6-member team using SolidWorks and Fusion 360 for CAD, subsystem iteration, and fabrication planning - Improved robot performance through 3D-printed component integration, geometric redesign, prototyping, and iterative testing - Managed subteam workflow, task allocation, and build schedules during competition preparation - Mentored newer members in CAD, fabrication, and system integration	

Research Assistant <i>Aerodynamic Testing and Experimental Analysis</i>	May 2025 – Jun. 2025 Walnut, CA
- Designed, assembled, and validated a vehicle-mounted aerodynamic test rig using CAD-designed mounts, 3D-printed components, and sensor instrumentation to characterize airfoil lift and drag under real driving conditions - Analyzed experimental data in MATLAB to evaluate aerodynamic performance trends across repeated test runs and improve measurement reliability - Refined structural components, instrumentation layout, and data-acquisition setup to improve test accuracy and repeatability - Integrated fabrication, instrumentation, and experimental validation to support data-driven aerodynamic design decisions	

PROJECTS

RC Vehicle Aerodynamics Testing & Optimization	ESP32, SolidWorks, 3D Printing
- Designed and built an RC car from scratch using SolidWorks, 3D-printed components, and an ESP32-based control system - Programmed the ESP32 and integrated it with the vehicle's electronic drive system - Applied Reynolds number, pressure drag, and boundary-layer analysis to refine body geometry and improve vehicle stability and aerodynamic performance - Incorporated an internal airflow channel inspired by Formula 1 F-duct concepts to reduce drag and improve overall vehicle performance	

BroncoHacks Posture Device	Python, Raspberry Pi, Sensors
- Built a posture-monitoring prototype using a Raspberry Pi, distance sensor, breadboarded circuitry, and a 3D-printed enclosure - Developed a Python-based interface to support device operation, sensor feedback, and user interaction - Integrated hardware and software under a 48-hour hackathon timeline to address a real-world posture problem	

Folding Knife Design & Fabrication	Magnacut, SolidWorks, Fabrication
- Designed and fabricated a folding knife using Magnacut steel and a back lock mechanism - Evaluated material behavior, load paths, and component tolerances through prototyping and mechanical design analysis to support strength, repeatability, and reliability - Applied fabrication and mechanical design principles to integrate structural performance with functional locking behavior	

UAS Drone Competition Project	SolidWorks, 3D Printing, Systems Engineering
- Served as systems engineer and 3D design engineer for a UAS competition team preparing for Cal State LA - Designed a payload drop mechanism to support mission-specific release operations - Applied CAD modeling, fabrication, and subsystem integration to develop a reliable mechanism for flight operations	

Raspberry Pi Sense HAT Controller	MATLAB, Raspberry Pi, Sense HAT
- Developed a MATLAB-based control interface for an 8x8 LED matrix using joystick and accelerometer inputs - Integrated sensor processing, conditional logic, and display control in an embedded mechatronic system	

SKILLS

Programming: MATLAB, Python, Arduino
CAD/Design: SolidWorks, Fusion 360
Hardware/Fabrication: Raspberry Pi, 3D Printing, Laser Cutting